PROJECT AND TEAM INFORMATION

Project Title

RakshaNetra - NHAI Worker Safety and Emergency Response System

Student / Team Information

Team Name: *Team-DSCPP-III-2026-T072*	*Tech Titans*
Team member 1 (Team Lead)	*Rawat, Soumya – 2510380027* *rawatsoumya01@gmail.com* 
Team member 2	*Anjum, Tazeen – 2510370373* *tazeenanjum142@gmail.com* 
Team member 3	*Chauhan, Sarthak Singh – 2510012694* *sarthakchauhanntt@gmail.com* 
Team member 4	*Pandey, Pranjal –2510370507* *pandeypranjal3107@gmail.com* 

PROPOSAL DESCRIPTION

Motivation

RakshaNetra is a safety management system for workers at NHAI construction sites.At a construction site,there are many workers and staff working,so managing their records manually becomes hard.The project will work as a simple software based system using C and C++. During an emergency, it is generally hard to quickly which person is facing and emergency or find the information we need at that time.

Our project will keep the data of workers and staff in one place. It will have information like their name,ID,role,and which work they are doing.By maintainig this record we will know whether a person is really an NHAI official worker or not.

Emergency situation is an important part of our project.If an accident happens,the staff can enter the details in the system. The system can then give the emergency a priority depending on how serious it is.It can also show which rescue teams are free and help select a team for the emergency.

We are using Data Structures in C and OOP concepts in C++ to make this project.We are using what we learn in class to make a system that can help with worker safety and emergencies.

We will create classes and objects for things like workers, staff, emergencies, and rescue teams.

OOP concepts such as encapsulation, inheritance, and polymorphism will be used wherever they are needed to keep the program simple and easy to handle.

The project will have a basic menu where the user can add new records, update existing ones, search for information, and view the stored details.

When an emergency is reported, it can be added to the system and given a priority based on how serious it is.

The system will also keep track of the rescue teams that are available and assign a suitable team to the emergency.

State of the Art / Current solution

At construction site, information such as worker's information, attendance,identity records,accident news,and safety information is usually managed by contractors.When an emergency happens there, the staff need to quickly know what has happened, which is involved,and what help is required.At present, this information may be written in registers or kept in different records. Because of this, finding the required details can take extra time during an emergency.

RakshaNetra is proposed to make this process easier by keeping the important safety information in one system.It will store details of workers and staff, check whether a person has official id,record emergency incidents,and help us identify which incidents need immediate attention. It will also help in selecting a suitable rescue team based on the situation.

The main idea of RakshaNetra is to keep the required information together so that staff do not have to search through different records during an emergency.

Project Goals and Milestones

The main goal of RakshaNetra : NHAI Worker Safety and Emergency Response System is to create a simple system that helps manage workers, staff, security, and emergencies at NHAI project sites.
Project Goals :

1. *1.Store and manage details of workers and staff.*

1. *2.Check the identity of workers and staff before allowing access.*

1. *3.Keep records of site entry and exit.*

1. *4.Record accidents and other emergency situations.*

1. *5.Give higher priority to serious emergencies.*

1. *6.Keep details of available rescue teams.*

1. *7.Help assign a suitable rescue team during an emergency.*

1. *8.Make worker and emergency information easy to find.*

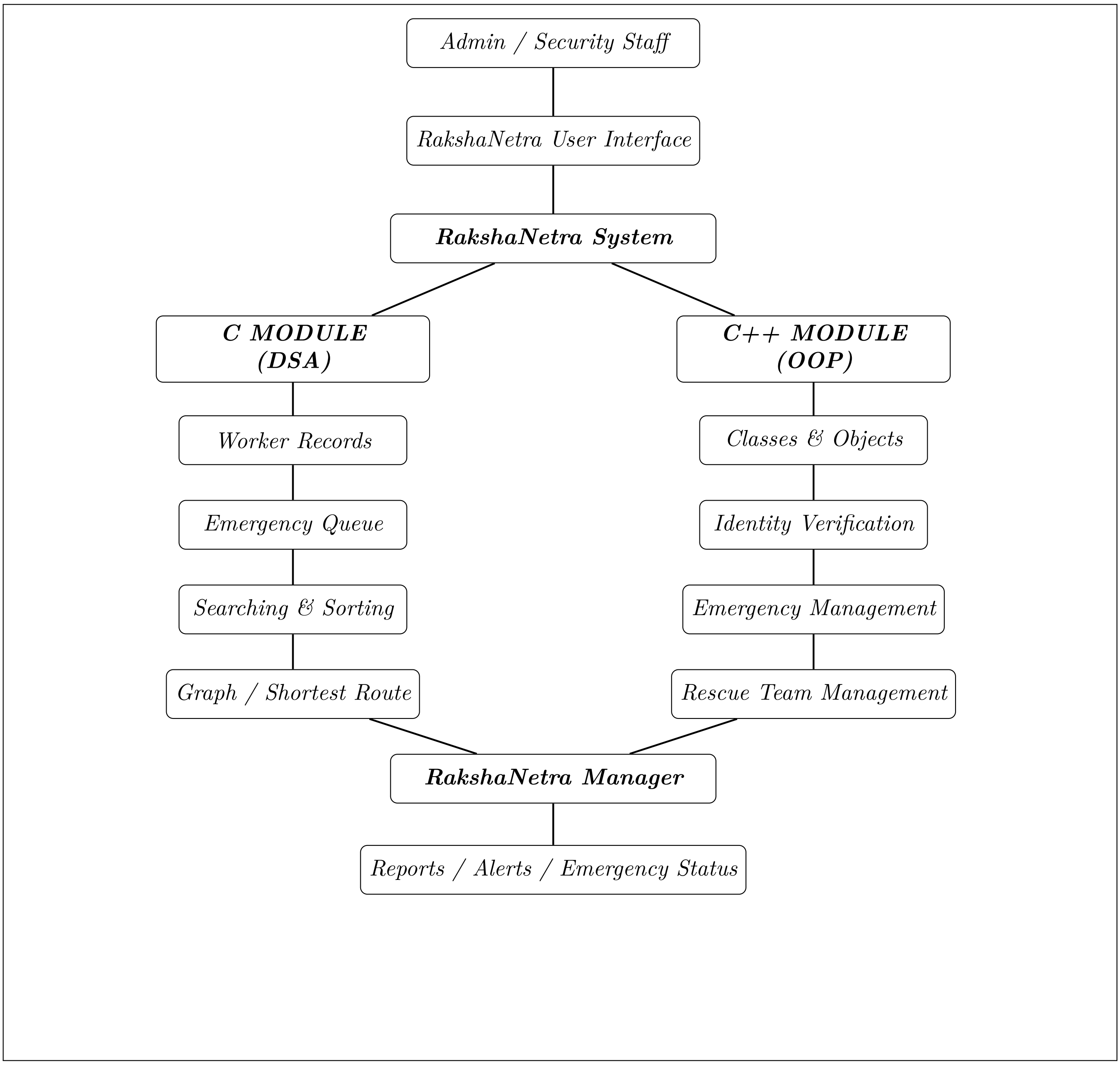
1. *9.Use DSA concepts in C and OOP concepts in C++ to build the system.*

Milestones :
The project will start with collecting the requirements and deciding the main features of the system.At construction site information such as worker details,attendance,identity,accident reports and safety information is managed by project teams or contractor.When an emergency happen the responsible staff have to identify the situation, check the available records, and arrange medical or rescue support. The problem is this information is present manually or in different records because of which finding the right information during an emergency can take time which can make the response slower.

Project Approach

The project will work as a software system using C and C++. Our project is to make a system which manage workers,staff,security details,emergences and rescue teams at the sites.
We will first know how the system should work making it cover Worker and Staff,Identity Verification,Security,Emergency and Rescue Team Management. Each part will have its own work, but all will be connected to each other.
For the Data Structures and Algorithms part, we will use C.When needed we can use arrays,linked lists,queues,priority queues and graphs.Searching and sorting will be useful for keeping the records when it is needed.
For an emergency we can represent the different places using a graph.The graph can be used to find a route for the rescue team to reach the location of the incident.
C++ will use Object-Oriented Programming concepts.Classes and objects will be created for workers,staff,emergencies,and rescue teams. Concepts such as encapsulation,inheritance, and polymorphism will be used.
The project will have a simple menu based dashboard so that users can easily add,update,search, and view records.Emergency cases can be registered and given priority according to their severity.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The main outcome of RakshaNetra : NHAI Worker Safety and Emergency Response System will be the following:

1. *1.Worker and staff record system which will keep the identification details, roles and basic information.*

1. *2.Verification system to identify the authorized workers and staff at construction site.*

1. *3.A safety monitoring module to record and manage the safety related activities and incidents.*

1. *4.An emergency management module to record the emergencies and classify based on severity.*

1. *5.A priority system which will help to identify the critical emergencies which will need faster attention.*

1. *6.A rescue team management feature to maintain the available rescue teams information and assist in assigning the suitable team during emergencies.*

1. *7.A simple C and C++ based implementation which will demonstrate the use of Data Structures and Object-Oriented Programming.*

The final system will provide an organized way to access the important safety information and support faster decision-making during emergencies.

Assumptions

We assume that the worker and staff details entered in the system are correct and updated when needed.Each worker will have a unique ID.Only authorized staff will add or change the records.We assume that emergency details such as the location of an accident will be entered correctly.Rescue team information will also be updated.The system works as a simple project model to demonstrate how worker safety and emergency information can be managed.

References

1. cppreference – C Language Reference — Used for C language syntax, arrays, structures, functions, pointers, memory management, and standard library concepts.
2. cppreference – C++ Language Reference — Used for classes, objects, inheritance, polymorphism, exception handling, and other C++ concepts
3. cppreference – C++ Standard Library — Used for STL containers, algorithms, iterators, and input/output facilities.
4. H. Yang et al., "Design and implementation of an identification system in construction site safety for proactive accident prevention," Accident Analysis and Prevention, 2012. — Used as a reference for worker identification, authorized personnel, and construction-site safety.
5.S. Rana, S. Sengupta, S. Jana, R. Dan, M. Sultana, and D. Sengupta, "Quick Accident Response System (QARS) Using Internet of Things (IoT)," 2020 Fifth International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN), Bangalore, India, 2020. — Used as a reference for accident reporting, emergency alerts, accident location, and emergency response.